Analysis of an ADC using a high-level model

The objective of this project is to develop a high-level model for a *nbit* SAR ADC. This model should accurately describe the behavior of the ADC described in the paper,
"A 6.94-fJ/Conversion-Step 12-bit 100-MS/s Asynchronous SAR ADC Exploiting Split-CDAC in 65-nm CMOS," in IEEE Access, vol. 9, pp. 77545-77554, 2021
Using this high-level model, it should be possible to obtain the digital output code of the ADC for any given input voltage. The model should include capacitor mismatch errors, parasitic capacitance errors and comparator offset errors.

a) Carefully read the paper to understand the operation of the ADC (Fig. 1). If necessary, obtain and read papers cited by this paper.
b) Analyze the capacitor array circuits (shown in Fig. 1 in the paper) to obtain the expression of the capacitor array voltages (V_{xp} and V_{xn}) as a function of Δv_{in}, the control bits and the capacitors values. Use the charge conservation principle in your analysis. Suggestion: first analyze the MSB sub-array separately.
c) Using the previous expression, write a Matlab/Octave model of the behavior of this SAR ADC, including the offset error of the comparator. Note that the SAR ADC uses two capacitor arrays (split into LSB and MSB arrays) and fully differential signals. Include random errors into the capacitors' values and into the comparator's offset voltage.
d) Use the previous model to calculate the transfer function of the ADC ($d_{out}(v_{in})$) and compute the transition voltages of the ADC for a given set of errors.
e) Use the model to measure the linearity of the ADC using INL, DNL and FFT.
f) Run Monte Carlo analysis of the ADC model to determine the sensitivity of the ADC to mismatch errors between the different components.
g) Compare your results with the results reported in the paper.
h) (optional) Estimate the minimum area and power dissipation of the DAC required for the ADC to have an expected linearity of 10.5 bits. Use the mismatch error estimation provided in Fig. 4 of the paper.

Information to include in the report:

- Explain why the authors decided to use a hybrid capacitor configuration and why they use the split capacitor array.
- Explain how you obtained the expressions and present the expressions in a readable format.
- Carefully explain all the relevant parts of your code, including how you obtained the values for the capacitors in the arrays.
- Demonstrate that your code is working correctly, by first presenting the INL and DNL for the case of zero errors in the circuit. Also present the evolution of Vxp and Vxn for each step for a given input voltage.
- Introduce specific errors into the values of key capacitors in the array (such as CB, Cdl, etc) and observe the resulting impact in the INL and DNL. Observe and explain the effect of the offset voltage on the ADC transfer function. Also compare the difference between using CB=2 or CB=16/15.
- Present a histogram of the linearity of the ADC resulting from the monte carlo analysis. Run this analysis for different magnitude of random errors in the capacitors. Present the best, worst and average INL and DNL from this analysis.
- When comparing your results with the paper note that the ADC in the paper has digital correction, which can result in a better performance.